MGU-UGP (HONOURS) REGULAR EXAMINATION
March 2026
Fourth SEMESTER
Discipline Specific Core Course (DSC) - MG4DSCMAT201
FUNDAMENTALS OF ANALYSIS
Scheme of Valuation

General Instructions for Valuation

- **Part A:** 5 questions to be answered. Total marks of each question should not exceed 2.
- **Part B:** 5 questions to be answered.Total marks of each question should not exceed 6.
- **Part C:** 3 questions to be answered. Total marks of each question should not exceed 10.
- Award full marks for a complete and correct alternate answer. Partial marks should be awarded for partially correct alternate answers based on the evaluator's discretion.

Part A: Short Answer Type Questions (2 marks each)

Answer any 5 questions.

1. **Exhibit a bijection between $\mathbb{N}$ and the set of all odd integers.** **[U] / [CO1]**

   Solution: The function $f : \mathbb{N} \to \{\text{odd integers}\}$ defined by $f(n) = 2n - 1$
   (Any valid bijection is acceptable). ... 2marks

2. **Determine the set A of all real numbers x such that $2x + 3 \leq 6$.** **[A] / [CO2]**

   Solution:
   $$2x + 3 \leq 6 \implies 2x \leq 3 \implies x \leq \frac{3}{2}.$$
   ... 2marks
   $A = \{x \in \mathbb{R} : x \leq 1.5\} = (-\infty, 1.5]$ ... 1mark.
   (Total Mark should not exceed 2)

3. **Give an example of a set that is bounded above, not below.** **[U] / [CO3]**

   Solution: The set $S = (-\infty, 0]$ is bounded above but not bounded below.
   (Any correct example is acceptable). ... 2marks

4. **Define a half-closed interval.** **[K] / [CO4]**

   Solution: A half-closed (or half-open) interval is an interval that includes one of its endpoints but not the other. ... 2marks
   Examples: $[1, 2)$ or $(2, 1]$. ... 1 mark
   (Total Mark should not exceed 2)

5. **Show that $e^{\bar{i}\theta} = e^{-i\theta}$.** **[U] / [CO5]**

   Solution: $e^{i\theta} = \cos\theta + i\sin\theta$ ... 1mark
   $(e^{\bar{i}\theta}) = \cos\theta - i\sin\theta = e^{-i\theta}$ ... 1mark

6. **Give an example of a set that is neither open nor closed in $\mathbb{C}$.** **[U] / [CO6]**

Solution: The set $S = \{z \in \mathbb{C} : 0 < |z| \leq 1\}$ is neither open nor closed.
(Any correct example is acceptable) ... 2marks.

7. **Evaluate** $\text{Log}(-1)$**.** **[A] / [CO8]**

Solution: The principal logarithm is $\text{Log}(z) = \ln|z| + i\,\text{Arg}(z)$. (Total Mark should not exceed 2) 1mark
For $z = -1$, $|-1| = 1$, $\text{Arg}(-1) = \pi$. ... 1mark
$\text{Log}(-1) = \ln 1 + i\pi = i\pi$. ... 1 mark
(Total Mark should not exceed 2)

8. **Find the value of** $z = x + iy$ **such that** $e^z = 1 + i$**.** **[An] / [CO8]**

Solution: $e^z = 1 + i = \sqrt{2}e^{i(\pi/4+2k\pi)}$. ... 1mark
Then $z = \ln(1+i) = \ln\sqrt{2} + i\left(\frac{\pi}{4} + 2k\pi\right)$, $k \in \mathbb{Z}$. ... 1mark
So $z = \frac{1}{2}\ln 2 + i\left(\frac{\pi}{4} + 2k\pi\right)$ ... 1mark
. (Total Mark should not exceed 2)

Part B: Short Essay Type Questions (6 marks each)

Answer any 5 questions.

9. **Show that** $1 < 2 < 3 < 4 < \ldots$ **[A]/[CO2]**

Solution: Since 1 is positive, i.e $0 < 1$. ... 2 Marks
Therfore by the theorem
" If $a > b$, then $a + c > b + c$"
, we have $0 + 1 < 1 + 1$
That is, $1 < 2$. ... 3 Marks
Similarly we have

$$1 + 1 < 2 + 1$$

That is

$$2 < 3$$

and so on ... 1 Mark

10. **Show that for all** $a, b \in \mathbb{R}$**,** $\min\{a, b\} = \frac{1}{2}(a + b - |a - b|)$**.** **[A]/[CO2]**

Solution: Consider two cases:
Case 1: $a \geq b$. Then $\min(a, b) = b$, and $|a - b| = a - b$. ... 1 mark
$\frac{1}{2}(a + b - |a - b|) = \frac{1}{2}(a + b - (a - b)) = \frac{1}{2}(2b) = b = \min(a, b)$. ... 3marks
Case 2: $a < b$. Then $\min(a, b) = a$ and $|a - b| = b - a$. ... 1 mark
$\frac{1}{2}(a + b - |a - b|) = \frac{1}{2}(a + b - (b - a)) = \frac{1}{2}(2a) = a = \min(a, b)$. ... 3marks
(Total Mark should not exceed 6)

11. **Let** S **be a nonempty subset of** $\mathbb{R}$ **that is bounded above and let** a **be any number in** $\mathbb{R}$**. Define the set** $a + S = \{a + s : s \in S\}$**. Prove that** $\sup(a + S) = a + \sup S$**.** **[U]/[CO3]**

Solution: Let $u = \sup S$. Then for all $s \in S$, $s \leq u$, so $a + s \leq a + u$. ... 2marks
Thus $a + u$ is an upper bound for $a + S$. ... 1 mark
Let v be any upper bound for $a + S$. Then $a + s \leq v$ for all s,
so $s \leq v - a$ for all s. ... 2marks
Thus $v - a$ is an upper bound for S, so $u \leq v - a$, i.e., $a + u \leq v$.
Hence $a + u$ is the least upper bound. ... 2 marks
(Total Mark should not exceed 6)

12. **Write the rational number represented by the periodic decimal** $7.31414\ldots14\ldots$**.[An]/[CO4]**

Solution: Let $x = 7.3141414\ldots$. and $1000x = 7314.141414\ldots$. ... 2marks
$1000x - 10x = 7314.1414\ldots - 73.1414\ldots = 7241$. ... 2marks
So $990x = 7241$, thus $x = \frac{7241}{990}$. ... 2marks

13. **Solve for the real and imaginary parts of the function (a) $f(z) = iz + 2$ and (b) $f(z) = x^2y + i(xy^2)$. In each part, check whether the Cauchy-Riemann equations are satisfied or not.** **[A]/[CO6]**

Solution:
(a) $f(z) = i(x + iy) + 2 = ix - y + 2 = (2 - y) + ix$.
So $u(x, y) = 2 - y$, $v(x, y) = x$. ... 1 mark
Then $u_x = 0$, $u_y = -1$, $v_x = 1$, $v_y = 0$.
$u_x = 0 = v_y$ and $u_y = -1 = -v_x$ So CR equations are satisfied. ... 2 marks
(b) $f(z) = x^2y + ixy^2$. Then $u = x^2y$, $v = xy^2$.
Then $u_x = 2xy$, $u_y = x^2$, $v_x = y^2$, $v_y = 2xy$. ... 1mark
$u_x = 2xy = v_y$ and $u_y = x^2, -v_x = -y^2 \implies$
$x^2 = -y^2$ which holds only if $x = y = 0$.
CR not satisfied except at origin. ... 2marks

14. **Determine whether the function $u(x, y) = (x^2 - y^2)^2$ is harmonic.** **[An]/[CO7]**

Solution: $u_x = 2(x^2 - y^2) \cdot 2x = 4x(x^2 - y^2)$. ... 1marks
$u_{xx} = 4(x^2 - y^2) + 4x(2x) = 12x^2 - 4y^2$. ... 2marks
$u_y = 2(x^2 - y^2) \cdot (-2y) = -4y(x^2 - y^2)$. ... 1mark
$u_{yy} = -4(x^2 - y^2) + (-4y)(-2y) = -4x^2 + 4y^2 + 8y^2 = -4x^2 + 12y^2$. ... 2marks
$u_{xx} + u_{yy} = (12x^2 - 4y^2) + (-4x^2 + 12y^2) = 8(x^2 + y^2)$,
which is not identically zero. So not harmonic. ... 1mark
(Total Mark should not exceed 6)

15. **Prove that $|\cos z| \geq |\cos x|$.** **[U]/[CO8]**

Solution: $\cos z = \cos(x + iy) = \cos x \cosh y - i \sin x \sinh y$. ... 1mark
Then $|\cos z|^2 = \cos^2 x \cosh^2 y + \sin^2 x \sinh^2 y$ ... 1mark
$= \cos^2 x(1 + \sinh^2 y) + \sin^2 x \sinh^2 y = \cos^2 x + \sinh^2 y$. 2marks
Since $\sinh^2 y \geq 0$, $|\cos z|^2 \geq \cos^2 x$, so $|\cos z| \geq |\cos x|$. ... 2marks

16. **Prove that $\cosh^2 z - \sinh^2 z = 1$.** **[A]/[CO8]**

Solution: Using definitions: $\cosh z = \frac{e^z + e^{-z}}{2}$, ... 1mark
$\sinh z = \frac{e^z - e^{-z}}{2}$. ... 1mark
Then $\cosh^2 z - \sinh^2 z = \left(\frac{e^z + e^{-z}}{2}\right)^2 - \left(\frac{e^z - e^{-z}}{2}\right)^2$ ... 2marks
$= \frac{(e^{2z} + 2 + e^{-2z}) - (e^{2z} - 2 + e^{-2z})}{4} = \frac{4}{4} = 1$. ... 2marks.

Part C: Essay Type Questions (10 marks each)

Answer any 3 questions.

17. **a) Give an example of a countable collection of finite sets whose union is not finite.**
b) Show that the union of two disjoint denumerable sets is denumerable. **[U]/[CO1]**

Solution:
a) Let $A_n = \{n\}$ for $n \in \mathbb{N}$.
Each A_n is finite, but $\bigcup_{n=1}^{\infty} A_n = \mathbb{N}$ is infinite (countably infinite).
or any correct example ... 3marks
b) Let $A = \{a_1, a_2, \dots\}$ and $B = \{b_1, b_2, \dots\}$ be disjoint denumerable sets. ... 1marks
Define $f : \mathbb{N} \to A \cup B$ by $f(2n - 1) = a_n$, $f(2n) = b_n$.
This is a bijection, so $A \cup B$ is denumerable.
(7 marks for clear bijection). ... 7marks
(Total Mark should not exceed 10)

18. **a) Let S be a nonempty subset of $\mathbb{R}$ that is bounded below. Prove that $\inf S = -\sup\{-s : s \in S\}$.**
b) Show that if a nonempty set S has a finite number of elements, then $\sup S \in S$.
[An]/[CO3]

Solution:

a) Let $T = \{-s : s \in S\}$. Since S is bounded below, T is bounded above. ...2marks
Let $\alpha = \inf S$. Then $\alpha \leq s$ for all s, so $-\alpha \geq -s$ for all s,
hence $-\alpha$ is an upper bound for T. ...1marks
For any upper bound M of T, $M \geq -s$ for all s,
so $-M \leq s$ for all s, hence $-M$ is a lower bound for S, so $-M \leq \alpha$, ...2marks
i.e., $M \geq -\alpha$.
Thus $-\alpha = \sup T$.
So $\inf S = \alpha = -\sup T = -\sup\{-s : s \in S\}$. ...2marks
b) For a finite set $S = \{x_1, x_2, \ldots, x_n\}$, let $M = \max\{x_1, \ldots, x_n\}$. ...3marks
By definition, M is an upper bound and $M \in S$,
hence M is the least upper bound, so $\sup S = M \in S$. ...2marks.
(Total Mark should not exceed 10)

19. **a) State and prove the density theorem.**
b) Can the same argument be said about irrationals? Justify your answer. **[A]/[CO3]**

Solution:

(a) i. *Statement:* If x and y are real numbers with $x < y$, then there exists a rational number r such that $a < r < b$. ...2 marks
ii. Proof of the case when $x = 0$ ...2 Marks
iii. Proof of the case when $x > 0$ ...3 Marks
iv. Proof of the case when $x < 0$ and $y > 0$ ...1 Mark
v. Proof of the case when $x < 0$ and $y < 0$ ...2 Marks

(b) Yes ...2 Marks
Since $x < y$, we have

$$\frac{x}{\sqrt{2}} < \frac{y}{\sqrt{2}}$$

By Density theorem there exists a non zero rational number r such that

$$\frac{x}{\sqrt{2}} < r < \frac{y}{\sqrt{2}}$$

Therefore

$$x < r\sqrt{2} < y$$

The number $r\sqrt{2}$ is the required irrational number ...2 Marks

(Total Mark should not exceed 10)

20. **Evaluate the following: a)** $\operatorname{Re}\frac{(2-3i)^2}{(2-3i)}$ **b)** $\operatorname{Im} z^3$ **c)** $\operatorname{Im}\frac{z}{\bar{z}}$ **[U]/[CO5]**

Solution:
a) $\frac{(2-3i)^2}{(2-3i)} = 2 - 3i$, so real part is 2. ...3marks
b) $z^3 = (x+iy)^3 = (x^3 - 3xy^2) + i(3x^2y - y^3)$. ...3marks
So $\operatorname{Im}(z^3) = 3x^2y - y^3$. ...1 mark
c) $\frac{z}{\bar{z}} = \frac{x+iy}{x-iy} = \frac{x^2-y^2+2ixy}{x^2+y^2}$. ...3marks
So $\operatorname{Im}\left(\frac{z}{\bar{z}}\right) = \frac{2xy}{x^2+y^2}$ for $z \neq 0$. ...1mark.
(Total Mark should not exceed 10)

21. **a) Find all the nth roots of z where $n = 1, 2, 3, \ldots$**
b) Prove $\log(z^{\frac{1}{n}}) = \frac{1}{n}\ln r + i(\frac{\Theta+2(pn+k)\pi}{n})$ **[E]/[CO8]**

Solution:
a) For $z = re^{i\theta}$, the nth roots are given by $z^{1/n} = r^{1/n}e^{i(\theta+2k\pi)/n}$,
$k = 0, 1, \ldots, n-1$. ...4marks
b) $\log w = \ln|w| + i\arg w$. ...2marks
With $z = re^{i\Theta}$, $z^{1/n} = r^{1/n}e^{i(\Theta+2p\pi)/n}$, ...2marks
so $\log(z^{1/n}) = \frac{1}{n}\ln r + i\left(\frac{\Theta+2p\pi}{n} + 2k\pi\right)$. ...4marks
(Total Mark should not exceed 10)

22. **a) Derive the logarithmic expression of** $\sin^{-1} z$**.**
b) Find $\frac{d}{dz}(\sin^{-1} z)$**.** **[An]/[CO8]**

Solution:

a) Let $w = \sin^{-1} z$, then $z = \sin w = \frac{e^{iw} - e^{-iw}}{2i}$. ...2marks

$2iz = e^{iw} - e^{-iw}$, so $e^{2iw} - 2ize^{iw} - 1 = 0$ ...2marks

Solve quadratic: $e^{iw} = iz \pm \sqrt{1 - z^2}$. Thus $w = -i \log(iz \pm \sqrt{1 - z^2})$. ...2marks

The principal value is $\sin^{-1} z = -i \log(iz + \sqrt{1 - z^2})$. ...1mark

b)Let $w = \sin^{-1} z$ and $\sin w = z$ 2marks

differentiate: $\cos w \frac{dw}{dz} = 1$, ...1marks

so $\frac{dw}{dz} = \frac{1}{\cos w} = \frac{1}{\sqrt{1 - \sin^2 w}} = \frac{1}{\sqrt{1 - z^2}}$. ...2 marks.

(Total Mark should not exceed 10)